

Sri Sivasubramaniya Nadar College of Engineering

5-Year Integrated M.Tech Computer Science & Engineering

CAT Assignment 1

Course Code: ICS1503 | Course Name: Web Application Development

Design Memory Card Matching Game using HTML, CSS, and JavaScript

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1. Website Overview

The Memory Card Matching Game is a browser-based game developed using HTML, CSS, and JavaScript. The game consists of 16 cards arranged in a 4 x 4 grid, representing 8 matching pairs of symbols. All cards start face down. The player flips two cards at a time by clicking on them. If the two selected cards match, they remain face up. If they do not match, they are flipped back after a delay of 1 second. The player wins the game once all 8 pairs have been successfully matched.

2. Technology Stack

- HTML5 - used for the structure and content of the web page.
- CSS3 - used for styling, layout (CSS Grid), and responsiveness.
- JavaScript (Vanilla) - used for game logic and interactivity.

3. Features Implemented

- 16 cards representing 8 matching pairs
- 4 x 4 grid layout using CSS Grid
- All cards initially displayed face down
- Random shuffling of cards at the start of every game
- Only two cards can be selected/flipped at a time
- Automatic match detection when two cards are flipped
- Matched cards remain face up and become unselectable
- Non-matching cards flip back automatically after 1 second
- The board is locked while two cards are being compared, preventing a third selection
- Already matched cards cannot be selected again
- A move counter tracks the number of attempts made

- Winning message displayed once all 8 pairs are matched
- Restart Game button to reset and reshuffle the board
- Responsive layout that adapts to smaller screens
- Basic accessibility support using semantic buttons, ARIA labels, and visible focus states

4. Folder Structure

```
bash:/mnt/d/Visual_Studio_Code/web-dev/sem_5/Assignments/CAT-1 Assignment$ tree
.
├── cat-assignment1.zip
├── index.html
├── report.pdf
├── script.js
└── style.css

1 directory, 5 files
bash:/mnt/d/Visual_Studio_Code/web-dev/sem_5/Assignments/CAT-1 Assignment$
```

index.html - Contains the structure of the Memory Card Game, including the heading, instructions, card grid, restart button, and status message.

style.css - Contains the layout, card styling (face-down, revealed, matched states), colors, hover/focus effects, and responsive design rules.

script.js - Contains the game logic, including card creation, shuffling, flipping, match checking, input locking, move counting, and restart functionality.

report.pdf - This documentation report describing the project.

5. Styling Summary

CSS Grid is used to arrange the 16 cards into a 4 x 4 layout. Each card has three distinct visual states: face-down (default dark background with no symbol shown), revealed/flipped (light background showing the symbol), and matched (green background indicating a completed pair). Spacing and alignment are handled through the grid gap property, and typography uses a simple, readable sans-serif font. A hover effect scales cards slightly to indicate interactivity, and a visible focus outline is provided for keyboard navigation. Media queries adjust the card and grid size for smaller screens, keeping the game usable on mobile devices.

6. How to Run

- 1 Extract the ZIP file.
- 2 Open the cat-assignment1 folder.
- 3 Open index.html in a web browser.
- 4 Click two cards at a time to flip them.
- 5 Match all 8 pairs to complete the game.
- 6 Use the Restart Game button to play again.

No server, installation, or external dependency is required to run this project. It works directly in any modern web browser.